

Problem Set I: Single and two qubit systems

Let $|i\rangle = [1, 0]^T$, $|j\rangle = [0, 1]^T$, $|m\rangle = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]^T$, $H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ and I_n the $n \times n$ identity matrix.

1. What is the vector obtained by projecting $|i\rangle$ along $|j\rangle$.
2. Find the 2×2 real matrices that projects an arbitrary given vector $|v\rangle$ along into the directions $|m\rangle$ and $|j\rangle$ respectively.
3. Let $|p\rangle$ be the vector obtained by projection $|i\rangle$ along $|m\rangle$. What is the projection of $|p\rangle$ along $|j\rangle$?
4. How will you explain the anomaly between the results of the first question and the third question?
5. Does H represent an observable?
6. What are the possible outcomes (real values) possible upon measuring $|i\rangle$ and $|m\rangle$ respectively with H ?
7. For each of the two vectors considered ($|i\rangle$ and $|m\rangle$), and for each of the real observable values, what is the probability of obtaining the value?
8. For each of the above four cases, what is the normalized post-measurement quantum state of the respective input qubit?
9. Does H represent a (theoretically) valid quantum gate?
10. Give an example of a 2×2 complex matrix that (theoretically) represents a valid quantum gate, but not an observable.
11. Give an example of a 2×2 complex matrix that (“theoretically” is to be assumed in all questions!) represents a valid observable, but not a quantum gate.
12. What is the mathematical condition for a 2×2 complex matrix to be a valid quantum gate as well as an observable?
13. Consider a two qubit quantum “circuit” with the H gate applied to the first qubit only, and no transformation applied to the second. What will be the equivalent 4×4 matrix on the whole two qbit system corresponding to this operation?
14. How will the vector $|m\rangle \otimes |i\rangle$ be transformed by the above circuit?
15. How will the vector $|i\rangle \otimes |m\rangle$ be transformed by the above circuit?
16. After transforming $|i\rangle \otimes |m\rangle$ with the above circuit, suppose we want to measure the first bit alone; what is the measurement operator (Hermitian 4×4 matrix)?
17. Corresponding to the measurement operator of the previous question, what are the various post-measurement states and the probability values of obtaining each post-measurement state?
18. Suppose in Question 13, the H gate was applied only on the second qubit (instead of the first), what will be the 4×4 matrix on the whole system? How will the answers to questions 14, 15, 16 and 17 change?
19. Is $H \otimes I_2$ a valid observable on the two qubit systems described in Question 13 and Question 18?
20. Upon measuring the system of Question 13 with $H \otimes I_2$, what are the possible post measurement states, measurement outcomes, and the probabilities of each of the outcomes, for input $|i\rangle \otimes |m\rangle$?
21. Upon measuring the system of Question 18 with $H \otimes I_2$, what are the possible post measurement states, measurement outcomes, and the probabilities of each of the outcomes, for input $|i\rangle \otimes |m\rangle$?
22. How will the answers to questions 20 and 21 change, if the measurement operator was $I_2 \otimes H$?
23. Is $H \otimes H$ a valid measurement operator on the two qubit system? If so, solve Questions 20 and 21 with the measurement operator $H \otimes H$.

When eigen value is not given, take computational basis and eigen values 1, and -1 so Pauli Z for a single bit